

A Project Report on
“ **SMART STUDY PLANNER (LMS)** ”

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ABSTRACT

The rapid growth of digital education has increased the need for efficient learning management systems that not only provide educational content but also help students manage their time and track their academic progress effectively. This project presents a web-based **Learning Management System (LMS) integrated with a Smart Study Planner**, designed to enhance the overall learning experience for both students and teachers.

The system enables teachers to create and manage courses, upload study materials, assign assignments, and design quizzes. Students can enroll in courses, access learning resources, submit assignments in the form of PDF or image files, and attempt quizzes to evaluate their understanding. One of the key features of the system is the **Smart Study Planner**, which automatically generates personalized study schedules for students based on assignment deadlines, pending tasks, and learning progress. This helps students utilize their time efficiently and avoid last-minute preparation.

In addition to planning, the system also provides an **Analytics Dashboard** that tracks student performance, including quiz scores, progress percentage, and weak subject identification. Based on this data, the system offers simple recommendations to improve learning outcomes. The integration of planning and analytics makes the platform more intelligent compared to traditional LMS solutions.

The project is developed using modern web technologies, with **React.js** for the frontend, **FastAPI** for the backend, and **MySQL** for database management. The system follows a modular architecture, ensuring scalability and ease of maintenance. The user interface is designed with a focus on simplicity, responsiveness, and professional appearance, supporting both light and dark themes.

Overall, the proposed system provides a structured and efficient approach to digital learning by combining course management, automated planning, performance tracking, and user-friendly design. It aims to improve student productivity, enhance learning consistency, and support teachers in managing academic activities effectively.

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INTRODUCTION

In recent years, the education sector has undergone a significant transformation with the adoption of digital technologies. Learning Management Systems (LMS) have become an essential part of modern education, enabling students and teachers to interact, share resources, and manage academic activities online. However, most traditional LMS platforms primarily focus on content delivery and lack intelligent features that help students plan their studies and monitor their performance effectively.

One of the major challenges faced by students is **time management and consistency in learning**. Students often struggle to balance multiple subjects, assignments, and deadlines, which leads to poor planning, last-minute preparation, and reduced academic performance. Additionally, existing systems do not provide sufficient insights into student progress or personalized guidance for improvement.

To address these issues, this project introduces a **Learning Platform with Smart Study Planner**, which combines the core functionalities of an LMS with intelligent planning and performance tracking features. The system allows teachers to create courses, upload assignments, and design quizzes, while students can enroll in courses, access study materials, submit assignments, and attempt quizzes.

The key highlight of this system is the **Smart Study Planner**, which automatically generates study schedules based on assignment deadlines, pending tasks, and student progress. This feature helps students organize their study routine efficiently and maintain consistency. Furthermore, the system includes an **Analytics Dashboard** that provides insights into student performance, such as quiz scores, progress levels, and identification of weak subjects. These insights help students focus on areas that need improvement.

The system is developed using **React.js** for the frontend, **FastAPI** for the backend, and **MySQL** as the database. It follows a modular and scalable architecture, ensuring efficient performance and easy maintenance. The user interface is designed to be modern, responsive, and user-friendly, supporting both light and dark themes

OBJECTIVE OF THE PROJECT

The primary objective of this project is to design and develop a **Learning Platform integrated with a Smart Study Planner** that improves the overall academic experience by combining learning, planning, and performance analysis into a single system. The detailed objectives of the project are as follows:

1. Development of a Web-Based Learning Management System (LMS)

The project aims to create a centralized web-based platform where teachers and students can interact effectively. Teachers can create courses, upload content, assign tasks, and monitor student activities, while students can access learning materials and participate in academic activities from anywhere.

2. Efficient Course Management and Enrollment

Another important objective is to provide a structured system for course management. Teachers can create courses with unique subject codes, and students can enroll using these codes. This ensures organized access to subject-specific content and simplifies the management of multiple courses.

3. Implementation of Smart Study Planner

A key objective of the project is to develop an intelligent **Smart Study Planner** that automatically generates study schedules. The planner considers assignment deadlines, number of pending topics, and available study time to create a personalized plan for each student. This helps in improving time management and reduces last-minute stress.

4. Assignment Creation and Submission System

The system is designed to allow teachers to upload assignments with deadlines and descriptions. Students can view these assignments and submit their work digitally in formats such as PDF or images. This eliminates manual submission and ensures better tracking of academic tasks.

5. Online Quiz System for Evaluation

The project includes an integrated quiz module where teachers can create multiple-choice questions. Students can attempt these quizzes to test their understanding. The system evaluates responses automatically and calculates scores, making the assessment process faster and more efficient.

6. Performance Tracking and Analytics

One of the major objectives is to provide an **Analytics Dashboard** that helps track student performance. The system analyzes quiz results and progress data to calculate average scores, identify weak subjects, and monitor learning trends. This enables students to focus on areas that require improvement.

7. Improvement of Time Management and Productivity

By combining planning and tracking features, the system aims to improve student productivity. The smart planner ensures that study tasks are distributed over time, helping students maintain consistency and avoid procrastination.

8. User-Friendly and Responsive Interface

The project focuses on developing a clean, modern, and easy-to-use interface. The UI is designed to be responsive so that it works smoothly on both desktop and mobile devices. It also supports both light and dark themes for better user experience.

9. Scalable and Modular System Design

The system is built using modern technologies such as **React (frontend), FastAPI (backend), and MySQL (database)**. The modular architecture ensures scalability, making it easier to add new features in the future without affecting existing functionality.

10. Providing Intelligent Recommendations

Although the system does not use advanced AI models, it provides basic intelligent suggestions based on user performance. For example, it identifies weak subjects and recommends more practice or study time for improvement.

11. Reduction of Manual Effort in Academic Management

The platform automates many academic processes such as assignment distribution, submission tracking, quiz evaluation, and study planning. This reduces manual effort for both teachers and students.

12. Enhancing Digital Learning Experience

The overall objective is to enhance the quality of digital education by integrating multiple features into a single platform. The system aims to make learning more structured, interactive, and efficient.

SCOPE OF THE PROJECT

The scope of this project is to develop a **web-based Learning Management System (LMS) integrated with a Smart Study Planner** that facilitates effective learning, planning, and performance tracking for both students and teachers. The system is designed to cover key academic activities in a digital environment while providing additional support through intelligent scheduling and analytics.

1. Academic Activity Management

The system provides a platform where teachers can create and manage courses, upload assignments, and design quizzes. Students can enroll in courses using subject codes and access all related academic content in one place. This ensures proper organization and easy accessibility of learning materials.

2. Smart Study Planning

A major scope of the project is the implementation of a **Smart Study Planner** that automatically generates study schedules for students. The planner uses assignment deadlines and pending tasks to distribute study workload over time. This helps students manage their time efficiently and maintain consistency in their studies.

3. Assignment and Submission System

The system allows teachers to assign tasks with deadlines and descriptions. Students can submit their work digitally in formats such as PDF or images. The system also maintains submission records, making it easier to track student progress and performance.

4. Online Quiz and Evaluation

The project includes a quiz module where teachers can create multiple-choice questions and students can attempt quizzes. The system evaluates answers automatically and generates scores, reducing manual effort and providing instant feedback.

5. Performance Analytics

The platform includes an **Analytics Dashboard** that tracks student performance based on quiz results and study activity. It helps identify weak subjects and provides basic recommendations to improve learning outcomes.

6. User Interface and Experience

The system is designed with a modern, responsive, and user-friendly interface. It supports both light and dark themes and ensures smooth usability across different devices such as desktops and mobile phones.

7. Limitations within Scope

The current system provides basic intelligent planning and analytics. Advanced AI-based personalization, real-time notifications, and mobile app integration are not included in the current scope but can be implemented in future versions.

SYSTEM OVERVIEW

The developed system is a **web-based Learning Platform with Smart Study Planner** that enables students and teachers to manage academic activities in a structured and digital manner. The system is designed to handle course management, assignment handling, quiz evaluation, study planning, and performance tracking through an interactive interface.

1. Working of the System

The system operates through a client-server architecture where the frontend (React) interacts with the backend (FastAPI) using API calls, and all data is stored in a MySQL database.

Users access the system through a web interface where they can perform actions based on their roles (student or teacher). The frontend sends requests to the backend, which processes them and retrieves or stores data in the database.

2. User Roles

The system supports two main types of users:

a) Teacher

Creates courses with subject codes

Uploads assignments with deadlines

Creates quizzes with multiple-choice questions

Views student submissions

b) Student

Enrolls in courses using subject codes

Views assignments and submits answers (PDF/Image)

Attempts quizzes and gets scores

Uses planner and smart planner for study scheduling

Views performance analytics

3. Core Functionalities Implemented

a) Course Management

Teachers create courses, and each course has a unique subject code. Students use this code to enroll and access course-related data.

b) Assignment System

Teachers upload assignments with descriptions and due dates. Students can:

View assignments

Upload solutions as files

Track submission status

c) Quiz System

Teachers create quizzes consisting of MCQ-type questions. Students:

Attempt quizzes

Select answers

Receive scores after submission

d) Manual Study Planner

Students can manually create study plans by selecting:

Course

Date
Topic
The planner tracks:
Pending tasks
Completed tasks
Expired tasks

e) Smart Study Planner

The system automatically generates study plans based on:
Assignment deadlines
Topics from assignment descriptions
It distributes study tasks over available days, helping students prepare systematically.

f) Analytics Dashboard

The system provides basic performance insights:
Average quiz score
Total quizzes attempted
Weak subject identification
This helps students understand their learning progress.

4. Data Handling

All data is stored and managed using a relational database. The system maintains records of:
Users
Courses
Assignments
Planner entries
Quiz questions and results
The backend ensures proper data validation and processing before storing or returning data.

5. User Interface

The system includes a modern and responsive UI with:
Dashboard view
Sidebar navigation
Dark/light theme support
Structured pages for each module
The interface is designed for ease of use and smooth interaction.

6. Key Characteristics of the System

Fully web-based system
Role-based access (student/teacher)
Integrated learning + planning + evaluation
Automated study scheduling (Smart Planner)
Real-time data interaction via APIs

TECHNOLOGY STACK

The project is developed using a modern and efficient technology stack that ensures scalability, performance, and ease of development. The system follows a full-stack architecture consisting of frontend, backend, and database technologies.

1. Frontend Technologies

React.js

- Used for building the user interface
- Provides component-based architecture
- Enables fast rendering using Virtual DOM
- Helps create a responsive and dynamic UI

HTML, CSS, JavaScript

- HTML is used for structuring web pages
- CSS is used for styling and layout design
- JavaScript is used for handling user interactions and logic

Custom CSS (theme.css)

- Used for designing modern UI
- Supports light and dark theme
- Provides responsive layout and animations

Axios

- Used for making API calls from frontend to backend
- Handles HTTP requests such as GET, POST, PUT, DELETE

2. Backend Technologies

FastAPI (Python Framework)

- Used to build RESTful APIs
- Provides high performance and fast response time
- Supports asynchronous programming
- Automatically generates API documentation (Swagger UI)

Python

- Core programming language used for backend logic
- Handles data processing, validation, and business logic

3. Database

MySQL

- Relational database used to store system data
- Stores data such as users, courses, assignments, planner, and quiz results
- Ensures data consistency and structured storage

4. Database Interaction

- SQLAlchemy (ORM)
- Used for database operations in Python
- Converts Python objects into database tables
- Simplifies CRUD operations

5. Development Tools

Visual Studio Code

- Used as the main code editor
- Supports extensions for React, Python, and debugging

Git & GitHub

- Used for version control and project management
- Helps in tracking changes and code backup

Postman / Swagger UI

- Used for testing APIs
- Helps verify backend endpoints and responses

6. Runtime Environment

Node.js

- Required to run React frontend
- Manages frontend dependencies using npm

Uvicorn

- ASGI server used to run FastAPI backend
- Handles API requests efficiently

7. Optional / Supporting Tools

- Browser (Chrome/Edge) for testing UI
- MySQL Workbench for database management
- File Upload Handling for assignment submissions

SYSTEM ARCHITECTURE

The **Learning Platform with Smart Study Planner** is designed using a **three-tier architecture**. This architecture divides the complete system into three major layers: **Frontend Layer**, **Backend Layer**, and **Database Layer**. Each layer has a separate responsibility and works together to provide a smooth learning experience for students, teachers, and admin users.

This architecture is suitable for the LMS project because the system includes many important modules such as user login, signup, dashboard, course management, assignments, quizzes, smart study planner, notifications, and progress tracking. By separating the system into layers, the project becomes easier to understand, maintain, test, and upgrade in the future.

1. Frontend Layer

The **Frontend Layer** is the presentation layer of the system. It is the part of the application that users directly interact with. In this project, the frontend is developed using **React.js**. React helps in creating a fast, dynamic, and user-friendly interface.

The frontend provides different interfaces for **students, teachers, and admin**. Each user gets a dashboard according to their role. For example, students can access courses, assignments, quizzes, and the smart study planner, while teachers can create courses, add assignments, and manage quizzes.

The frontend does not directly communicate with the database. It sends requests to the backend through APIs. After receiving the response from the backend, it displays the result to the user in a proper format.

Main functions of the frontend layer are:

- Display login and signup pages
- Provide role-based dashboards
- Show course details and study materials
- Display assignments and quizzes
- Provide smart study planner interface
- Show generated study plans
- Display notifications and messages
- Show student progress and activity status
- Send user input to backend APIs
- Display backend responses such as success or error messages

For example, when a student enters login details, the React frontend sends the email and password to the backend. After successful verification, the student is redirected to the student dashboard.

2. Backend Layer

The **Backend Layer** is the application layer of the system. It is developed using **FastAPI**. This layer handles the main logic and processing of the LMS project. It receives requests from the frontend, validates the input, communicates with the database, and sends responses back to the frontend.

The backend acts as a bridge between the frontend and database. It ensures that the frontend cannot directly access the database, which improves security and control.

The backend manages important operations such as authentication, role-based access, course handling, assignment management, quiz management, study planner generation, notification management, and error handling.

Main functions of the backend layer are:

- Handle API requests from frontend
- Validate user input
- Manage student, teacher, and admin authentication
- Provide role-based access control
- Manage courses and study materials
- Manage assignments and submissions
- Manage quizzes and answers
- Generate smart study plans
- Store and retrieve notifications
- Communicate with MySQL database
- Send structured responses to frontend
- Handle errors and exceptions

For example, when a teacher creates an assignment, the frontend sends assignment details to the backend. The backend checks the data, stores the assignment in the database, and sends a success message to the frontend.

3. Database Layer

The **Database Layer** is responsible for storing all permanent data of the LMS project. In this system, **MySQL** is used as the database. MySQL stores all important records such as user details, courses, assignments, quizzes, study plans, notifications, and progress details.

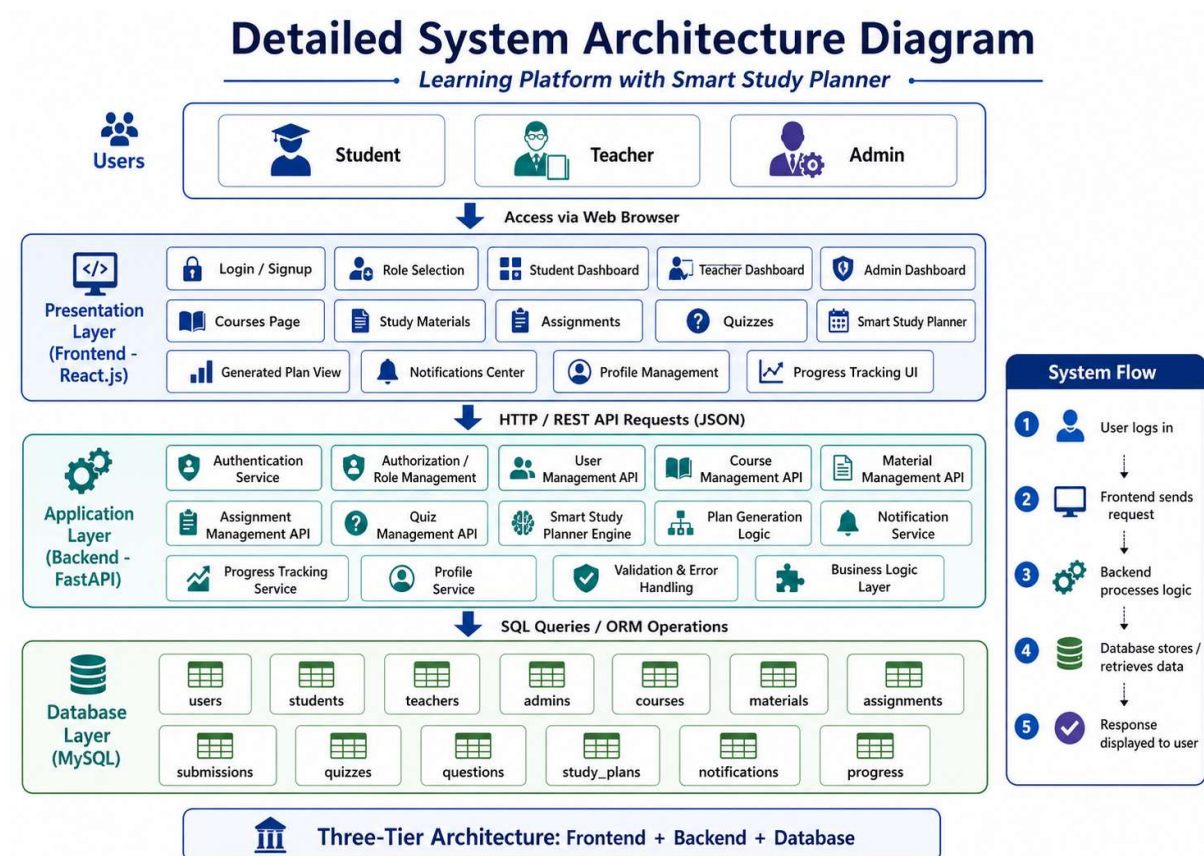
The database does not communicate directly with the frontend. It only communicates with the backend. This makes the system more secure and organized.

Main data stored in the database includes:

- Student details
- Teacher details
- Admin details
- Login and signup records

- Course details
- Study material information
- Assignment details
- Quiz questions and options
- Student submissions
- Generated study plans
- Notifications
- Progress records

For example, when a student opens the dashboard, the backend fetches the student's course details, assignments, quizzes, notifications, and study plan data from the MySQL database and sends it to the frontend.



SYSTEM CONFIGURATION

The System Configuration describes the hardware and software requirements needed to develop, run, and use the Learning Platform with Smart Study Planner. Since this project is a web-based LMS, it requires a proper frontend environment, backend environment, database server, and web browser for smooth execution.

1. Hardware Configuration

Minimum Hardware Requirements

Component	Requirement
Processor	Intel Core i3 or equivalent
RAM	4 GB minimum
Storage	10 GB free space
Display	1366 × 768 resolution
Network	Internet connection / Local network
Input Devices	Keyboard and Mouse

Recommended Hardware Requirements

Component	Requirement
Processor	Intel Core i5 or higher
RAM	8 GB or higher
Storage	20 GB free space or more
Display	Full HD display
Network	Stable internet connection
Input Devices	Keyboard and Mouse

2. Software Configuration

Software	Purpose
Windows 10 / Windows 11	Operating system
Visual Studio Code	Code editor
Python 3.x	Backend development

Software	Purpose
FastAPI	Backend framework
Uvicorn	ASGI server to run FastAPI
React.js	Frontend development
Node.js	Required to run React
npm	Package manager for frontend
MySQL Server	Database management
MySQL Workbench	Database design and management
Web Browser	To access the LMS application
Git	Version control
Docker	Containerization and deployment

3. Frontend Configuration

The frontend of the system is developed using React.js. It provides the user interface for students, teachers, and admin users.

Frontend Requirements

Tool / Technology	Description
React.js	Used to build the user interface
Node.js	Required to run React project
npm	Used to install frontend packages
HTML	Used for page structure
CSS	Used for styling and design
JavaScript	Used for frontend logic
Axios / Fetch API	Used to connect frontend with backend APIs

The frontend runs in the browser and communicates with the backend using REST API requests.

4. Backend Configuration

The backend is developed using FastAPI. It handles authentication, role management, courses, assignments, quizzes, study planner, notifications, and database operations.

Backend Requirements

Tool / Technology	Description
Python 3.x	Main programming language
FastAPI	Backend framework
Uvicorn	Server for running FastAPI application
SQLAlchemy	ORM for database operations
PyMySQL	Connector for MySQL database
Pydantic	Data validation
CORS Middleware	Allows frontend-backend communication

The backend receives requests from the frontend, processes logic, connects with MySQL, and sends responses back to the frontend.

5. Database Configuration

The project uses MySQL as the database. It stores all important data of the LMS project.

Database Requirements

Tool / Technology	Description
MySQL Server	Stores project data
MySQL Workbench	Database management tool
SQLAlchemy ORM	Connects backend with database
Database Name	LMS

DATABASE DESIGN

The **Database Design** of the **Learning Platform with Smart Study Planner** defines how the data of students, teachers, admin, courses, assignments, quizzes, study plans, notifications, and progress records is stored and managed. The project uses **MySQL** as the database because it is reliable, structured, and suitable for web-based applications.

The database is designed in a relational format where data is stored in different tables. Each table contains related information, and tables are connected using primary keys and foreign keys.

The database of the Learning Platform with Smart Study Planner is designed using a relational database model. The project uses MySQL to store and manage all important data related to students, teachers, admin, courses, assignments, quizzes, study plans, notifications, and progress tracking.

The main database name is `lms_db`. It contains multiple tables such as users, students, teachers, admins, courses, materials, assignments, submissions, quizzes, questions, study_plans, notifications, and progress.

The users table stores common login details like name, email, password, and role. Based on the role, user information is connected with separate tables such as students, teachers, and admins. This helps the system provide role-based access and dashboards.

The courses table stores course details created by teachers. Study materials, assignments, and quizzes are connected with courses so that students can easily access learning content. The assignments and submissions tables manage assignment creation and student submissions, while the quizzes and questions tables manage quiz details and multiple-choice questions.

The study_plans table stores smart study plans generated for students. It includes subjects, topics, study date, time, duration, priority, and completion status. The notifications table stores system messages such as assignment updates, quiz alerts, course updates, and planner reminders. The progress table stores student learning progress, including completed topics, assignment status, quiz score, and progress percentage.

Foreign keys are used to connect related tables. For example, a teacher can create many courses, a course can have many assignments and quizzes, and a student can have many study plans and submissions. This structure reduces data duplication and keeps the database organized.

Overall, the database design supports all major features of the LMS system, including login/signup, role-based access, course management, study materials, assignments, quizzes, smart study planner, notifications, and progress tracking. It makes the system secure, structured, scalable, and easy to maintain.

WITH EXPLANATION I POINTS OF TABLES

Database Summary with Explanation of Tables

The database of the Learning Platform with Smart Study Planner is designed using MySQL. It stores all important information related to users, courses, study materials, assignments, quizzes, smart study plans, notifications, and progress tracking. The database is divided into multiple tables so that data can be stored in an organized and secure way.

Main Tables and Their Explanation

1. Users Table

The users table stores the common login information of all users.

It stores:

User ID

Name

Email

Password

Role

Explanation:

This table is used for login and signup. It identifies whether the user is a student, teacher, or admin. Based on the role, the system redirects the user to the correct dashboard.

2. Students Table

The students table stores student-specific details.

It stores:

Student ID

User ID

Semester

Department

Course enrolled

Explanation:

This table is connected with the users table. It stores extra details about students and helps the system manage student activities such as course access, assignment submission, quiz attempts, and study planner usage.

3. Teachers Table

The teachers table stores teacher-specific details.

It stores:

Teacher ID

User ID

Department

Subject specialization

Explanation:

This table stores teacher information. Teachers can create courses, upload study materials, add assignments, and create quizzes. It helps connect teachers with the courses and content they manage.

4. Admins Table

The admins table stores admin-related information.

It stores:

Admin ID

User ID

Admin code

Explanation:

This table is used to identify admin users. Admin users can manage the overall LMS system, monitor users, and control platform-level activities.

5. Courses Table

The courses table stores course details created by teachers.

It stores:

Course ID

Course title

Course description

Teacher ID

Created date

Explanation:

This table manages all courses available in the LMS. Each course is connected to a teacher. Students can view and access course-related content from this table.

6. Materials Table

The materials table stores study material uploaded by teachers.

It stores:

Material ID

Course ID

Material title

File link or file path

Upload date

Explanation:

This table stores notes, PDFs, documents, and learning resources related to a course. Students can access study materials from their course page.

7. Assignments Table

The assignments table stores assignment details.

It stores:

Assignment ID

Course ID

Teacher ID

Assignment title

Description

Deadline

Marks

Explanation:

Teachers use this table to create assignments for students. It stores assignment instructions, deadlines, and marks. Students can view assignments through their dashboard.

8. Submissions Table

The submissions table stores student assignment submissions.

It stores:

Submission ID

Assignment ID

Student ID

Submitted file link

Submission date

Status

Marks obtained

Explanation:

This table tracks which student submitted which assignment. It also stores submission status such as submitted, late, or checked. Teachers can use this table to review submissions and assign marks.

9. Quizzes Table

The quizzes table stores quiz information.

It stores:

Quiz ID

Course ID

Teacher ID

Quiz title

Total marks

Created date

Explanation:

Teachers can create quizzes for specific courses. This table stores quiz-level information such as title, course, teacher, and total marks.

10. Questions Table

The questions table stores questions related to quizzes.

It stores:

Question ID

Quiz ID

Question text

Option A

Option B

Option C

Option D

Correct option

Explanation:

This table stores multiple-choice questions for quizzes. Each question is connected with a quiz. Students answer these questions while attempting quizzes.

11. Study Plans Table

The study_plans table stores smart study plans generated for students.

It stores:

Plan ID

Student ID

Subject

Topic

Study date

Study time

Duration

Priority

Status

Explanation:

This is one of the most important tables of the project. It stores daily or weekly study plans generated by the Smart Study Planner. It helps students manage their study schedule and track whether a task is pending or completed.

12. Notifications Table

The notifications table stores important messages and alerts.

It stores:

Notification ID

User ID

Message

Type

Read status

Created date

Explanation:

This table is used for the notification center. It stores alerts related to assignments, quizzes, courses, study plans, and general updates. Users can see these notifications in their dashboard.

13. Progress Table

The progress table stores student learning progress.

It stores:

Progress ID

Student ID

Course ID

Completed topics

Total topics

Assignment status

Quiz score

Progress percentage

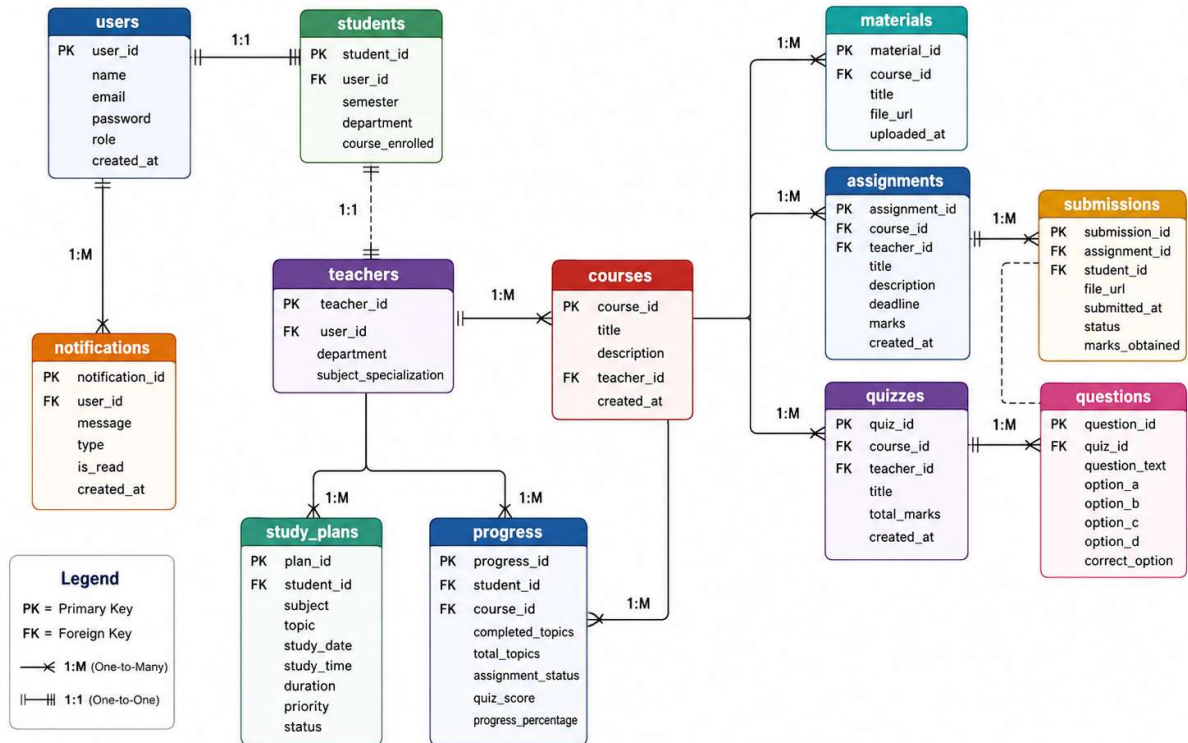
Explanation:

This table helps track student performance and learning progress. It shows how much course content a student has completed, assignment status, quiz score, and overall progress percentage.

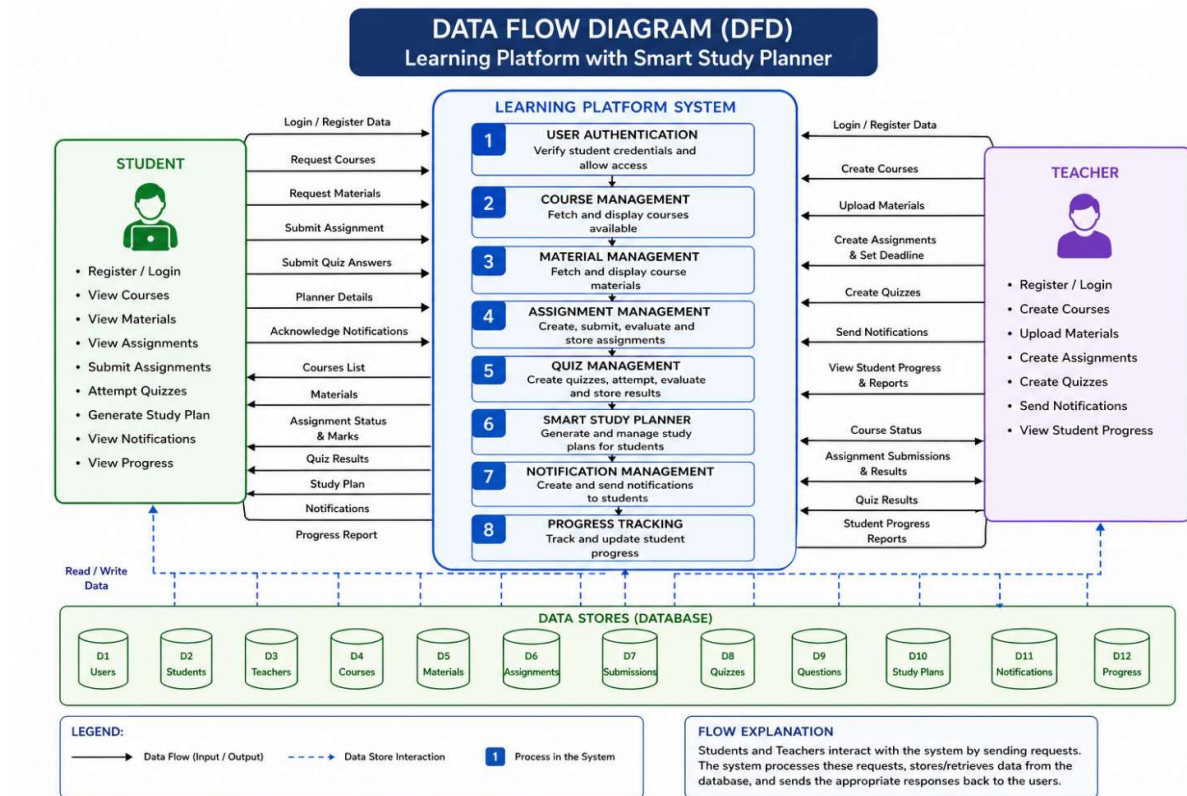
ER DIAGRAM

ER Diagram – Learning Platform with Smart Study Planner

Database design with only Student and Teacher roles



DFD DIAGRAM



CONCLUSION

The Learning Platform with Smart Study Planner is a web-based LMS designed to support both students and teachers in the learning process. The system provides an organized platform where teachers can manage courses, upload study materials, create assignments, and prepare quizzes, while students can access learning resources, submit assignments, attempt quizzes, generate study plans, and track their progress.

The project is developed using React.js for the frontend, FastAPI for the backend, and MySQL for database management. This technology combination makes the system fast, reliable, scalable, and easy to maintain. The use of a three-tier architecture separates the frontend, backend, and database layers, which improves system performance and security.

One of the most important features of this project is the Smart Study Planner. It helps students manage their study time by generating planned schedules based on subjects, topics, time, duration, and priority. This feature improves time management and helps students complete their learning tasks in a systematic way.

Overall, the project successfully provides a digital learning environment that reduces manual work, improves communication between teachers and students, and supports better academic planning. The system is useful for educational institutions because it combines course management, assignments, quizzes, notifications, progress tracking, and smart planning in a single platform. In the future, the project can be enhanced with features like video lectures, attendance tracking, AI-based recommendations, certificate generation, chat support, and mobile application support.